

TMUA Mock Paper 1

(By Zetta)

75 Minutes, No calculator
Answer key on final page

1 Given that

$$\int_1^a (ax + 2) dx = \frac{5\sqrt{2}}{2} - 2$$

Find the value of a

- A** 2
- B** $\sqrt{2}$
- C** $\sqrt{2} + 1$
- D** $\sqrt{2} - 1$
- E** 1.5

2 Given that the points $(-2, 0)$, $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$ and $(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$ all lie on the circle C with radius r and center $(a, 0)$

Find $r - a$

A 1

B $1 - \sqrt{2}$

C 2

D $3 + \sqrt{3}$

E 3

3 What are the solution(s) to

$$\ln(x^2 + 4x - 90) = \ln(-2x + 1)$$

For all real x ?

- A** $13, 7$
- B** $-13, 7$
- C** $-13, -7$
- D** -13
- E** 7

4 You are given that

$$p(x) = -6 + x + x^2 + x^3 + x^4 + x^5 + \dots$$

Find the value of x such that $p(x) = 0$

- A** $-\frac{1}{7}$
- B** $-\frac{1}{6}$
- C** $\frac{8}{7}$
- D** $\frac{1}{7}$
- E** $\frac{6}{7}$

5 After Euclid High School's last basketball game, it was determined that $\frac{1}{4}$ of the team's points were scored by Alexa and $\frac{2}{7}$ were scored by Brittany. Chelsea scored 15 points. None of the other 7 team members scored more than 2 points. What was the total number of points scored by the other 7 team members?

- A** 10
- B** 11
- C** 12
- D** 13
- E** 14

6 Evaluate

$$\sum_{n=1}^{6000} \frac{1}{n(n+1)}$$

A $\frac{6000}{6001}$

B $\frac{6001}{6000}$

C $\frac{5999}{6000}$

D $\frac{6000}{5999}$

E $\frac{5999}{6001}$

7 Let $A(n)$ be the area of a regular polygon with n sides, each of length 1, where $n \geq 3$. Find, in terms of n , an expression for

$$\frac{A(n)}{A(n+1)}$$

[Angles in degrees]

A $\frac{(n+1) \tan(90 - \frac{180}{n+1})}{n \tan(90 - \frac{180}{n})}$

B $\frac{n \tan(90 - \frac{180}{n})}{(n+1) \tan(90 - \frac{180}{n+1})}$

C $\frac{n \tan(180 - \frac{360}{n})}{(n+1) \tan(180 - \frac{360}{n+1})}$

D $\frac{4n \tan(180 - \frac{360}{4n})}{(n+1) \tan(180 - \frac{360}{n+1})}$

E $\frac{n \tan(\frac{180}{n})}{(n+1) \tan(\frac{180}{n+1})}$

F $\frac{n \tan(180 - \frac{360}{n})}{n \tan(90 - \frac{180}{n})}$

8 Which of the following expressions is closest to the value of

$$\sqrt{100001} - \sqrt{100000}$$

A $\frac{1}{\sqrt{100000}}$

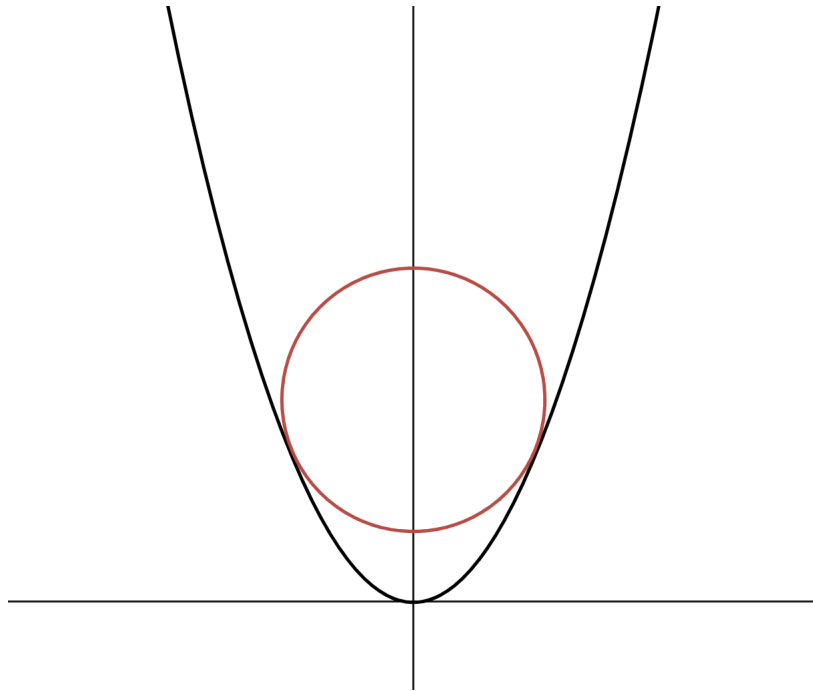
B $\frac{1}{2\sqrt{100000}}$

C $\frac{1}{10\sqrt{100000}}$

D $\frac{1}{100\sqrt{100000}}$

E $\frac{1}{100000\sqrt{100000}}$

- 9** A circle of radius r is *dropped* into the curve $y = x^2$ such that it only touches the curve.



The circle's center is $(0, c)$
Express c in terms of r .

- A** $\frac{1 + 2r^2}{2}$
B $\frac{1 - 4r^2}{4}$
C $\frac{4r}{r^2 - 2}$
D $\frac{r^2}{4} - 1$
E $\frac{1 + 4r^2}{4}$
F $\frac{1 - 6r^2}{2}$
G $\frac{r^3 - r}{4}$

10 Simplify

$$\sqrt{3 - 2\sqrt{2}}$$

A $\sqrt{2}$

B $2 + \sqrt{2}$

C $\sqrt{2} - 1$

D $2 - \sqrt{2}$

E $1 - \sqrt{2}$

F $\frac{\sqrt{2}}{2} + \frac{1}{2}$

G $3 - 3\sqrt{2}$

11 Given that $a \neq d \neq 0$ and that

$$\frac{a^2 - 2a + d^2}{2d + 2ad - 1} = 1$$

What is the value of $a + d$

- A** 1
- B** 2
- C** 3
- D** 4
- E** $\sqrt{2}$
- F** $\frac{1}{2}$
- G** -1

12 Xander rolls 3 fair six-sided dice. Given that at least 2 of them show a 3, what is the probability the third also shows a 3?

A $\frac{3}{8}$

B $\frac{3}{18}$

C $\frac{1}{3}$

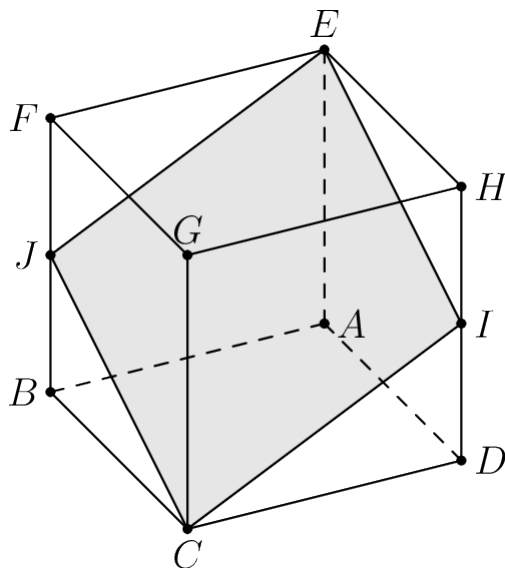
D $\frac{1}{18}$

E $\frac{1}{6}$

F $\frac{1}{2}$

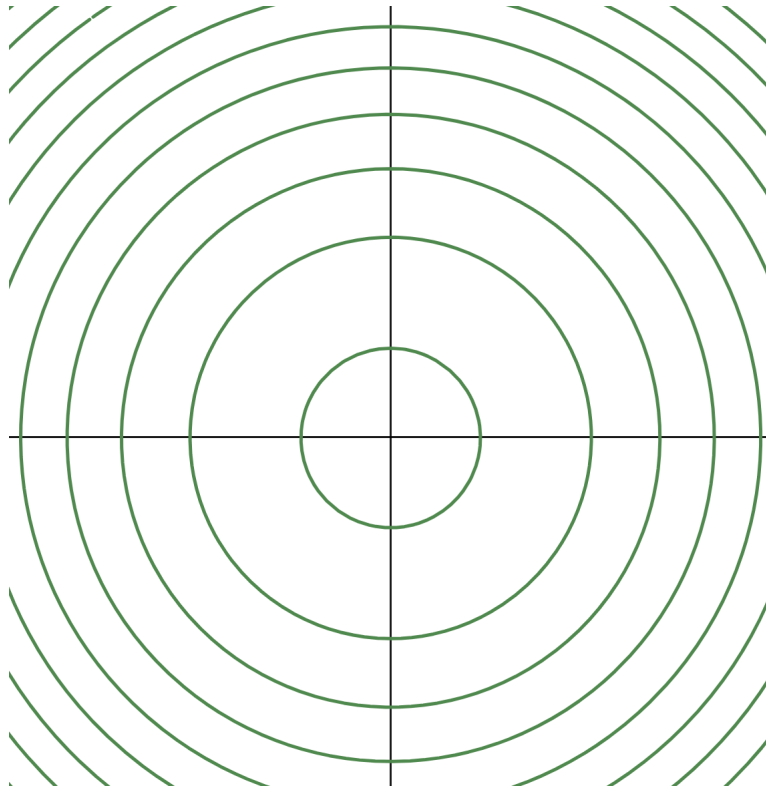
G $\frac{1}{16}$

13 In the cube $ABCDEFGH$ with opposite vertices C and E , J and I are the midpoints of segments $\overline{FB}$ and $\overline{HD}$, respectively. Let R be the ratio of the area of the cross-section $EJCI$ to the area of one of the faces of the cube. What is R^2 ?



- A** $\frac{5}{4}$
- B** $\frac{4}{3}$
- C** $\frac{3}{2}$
- D** $\frac{25}{16}$
- E** $\frac{9}{4}$
- F** $\frac{7}{4}$

14 The graph of $\tan(x^2 + y^2) = 1$ is shown below (angles in radians).



The graph consists of a central circle surrounded by infinitely many **annuluses** or rings.

The central circle has area C and each annulus has the same constant area A .

What is $\frac{A}{C}$?

[An *annulus* is the region between two concentric circles (like a ring)]

- A 4
- B $\frac{1}{2}$
- C π^2
- D $\frac{3}{4\pi}$
- E $\frac{\pi^2 + 1}{4\pi}$
- F $\frac{1}{\pi}$

15 Given that x is in radians, write

$$\sin(2 \arctan(x))$$

without using trigonometric functions.

A $\frac{1}{\sqrt{1+x^2}}$

B $\frac{1+x^2}{\sqrt{2x}}$

C $\frac{x}{\sqrt{1+x^2}}$

D $\frac{2x}{1+x^2}$

E $\frac{1}{\sqrt{2x}}$

F $\frac{2x}{1-x^2}$

16 a , b and c are positive integers so that

$$\frac{\left(\frac{a}{c} + \frac{a}{b} + 1\right)}{\left(\frac{b}{a} + \frac{b}{c} + 1\right)} = 17$$

How many solutions are there to the inequality

$$a + 2b + 3c \leq 50$$

A 14

B 15

C 16

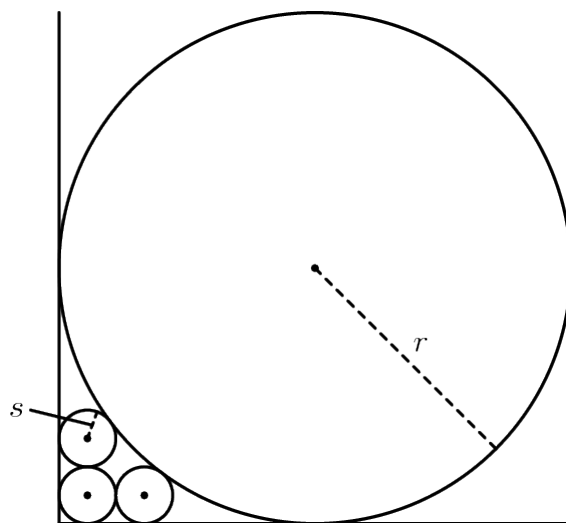
D 17

E 18

F 19

G 0

17 Three circles of radius s are drawn in the first quadrant of the xy -plane. The first circle is tangent to both axes, the second is tangent to the first circle and the x -axis, and the third is tangent to the first circle and the y -axis. A circle of radius $r > s$ is tangent to both axes and to the second and third circles. What is r/s ?



- A** 5
- B** 6
- C** 8
- D** 9
- E** 10
- F** 12

18 Let S be the set of all points with coordinates (x, y, z) , where x , y , and z are each chosen from the set $\{0, 1, 2\}$. How many equilateral triangles all have their vertices in S ?

- A** 72
- B** 80
- C** 86
- D** 94
- E** 96
- F** 102

19 Find the number of integer values of k in the range $-500 \leq k \leq 500$ for which the equation $\log(kx) = 2\log(x+2)$ has exactly one real solution.

A 499

B 500

C 501

D 502

20 Given that:

$$f(x) = x^{2008} - 2x^{2007} + 3x^{2006} - 4x^{2005} + \cdots + 2007x^2 - 2008x + 2009$$

and that:

$$g(x) = \frac{2012x + 2009}{(1 + x^2)}$$

Which of the following is a solution to $g(x) = f(x)$

A 1

B $2^{\frac{1}{2009}}$

C $3^{\frac{1}{2009}}$

D $4^{\frac{1}{2009}}$

E $2^{\frac{1}{2008}}$

F $3^{\frac{1}{2008}}$

F $4^{\frac{1}{2008}}$

Answer Key

1	B
2	C
3	D
4	E
5	B
6	A
7	B
8	B
9	E
10	C
11	B
12	G
13	C
14	A
15	D
16	A
17	D
18	B
19	C
20	B

This paper is designed to be very hard, with many questions possibly being similar to or harder than the later TMUA questions.

If you are looking for worked solutions try searching up the questions, multiple were taken from the internet, particularly SMC Team challenge and AMC 8/AMC 12

Note: there may be errors